

CONCEPT OF HYDROLOGY

- Water is vital for all living organisms on Earth.
- For centuries, people have been investigating where water comes from and where it goes, why some of it is salty and some is fresh, why sometimes there is not enough and sometimes too much. All questions and answers related to water have been grouped together into a discipline.
- The name of the discipline is **HYDROLOGY** and is formed by two Greek words: "hydro" and "logos" meaning "water" and "science".



CONCEPT OF HYDROLOGY

Hydrology is the science that deals with the waters of the Earth, their occurrence, circulation and distribution, their chemical and biological properties and their reaction with their environment, including their relation to living things.

Most hydrologic problems are related to either quantity or quality of water or both.

Interrelationship of geologic materials and processes with water is [hydrogeology](#). Typical definition of hydrogeology is occurrence, distribution, movement and geological interaction of water in the earth surface crust.

Geohydrology is sometimes used as synonym for hydrogeology, although it more properly describes an engineering field dealing with subsurface fluid hydrology

Similarly study of glaciers is [glaciology](#), of inland waters is [limnology](#) and distribution of oceans is [oceanography](#)

The collective mass of water found on, under, and over the surface of a planet is called the [hydrosphere](#).

Earth's approximate water volume (the total water supply of the world) is 1,360,000,000 km³ (326,000,000 mi³).

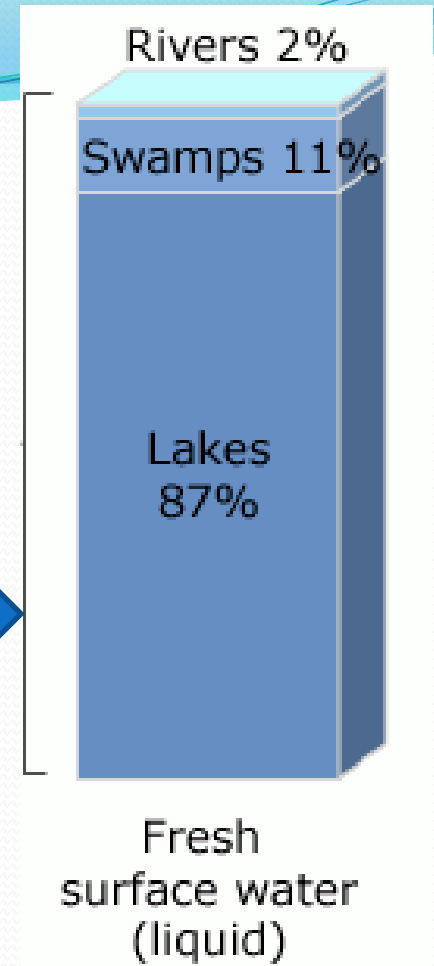
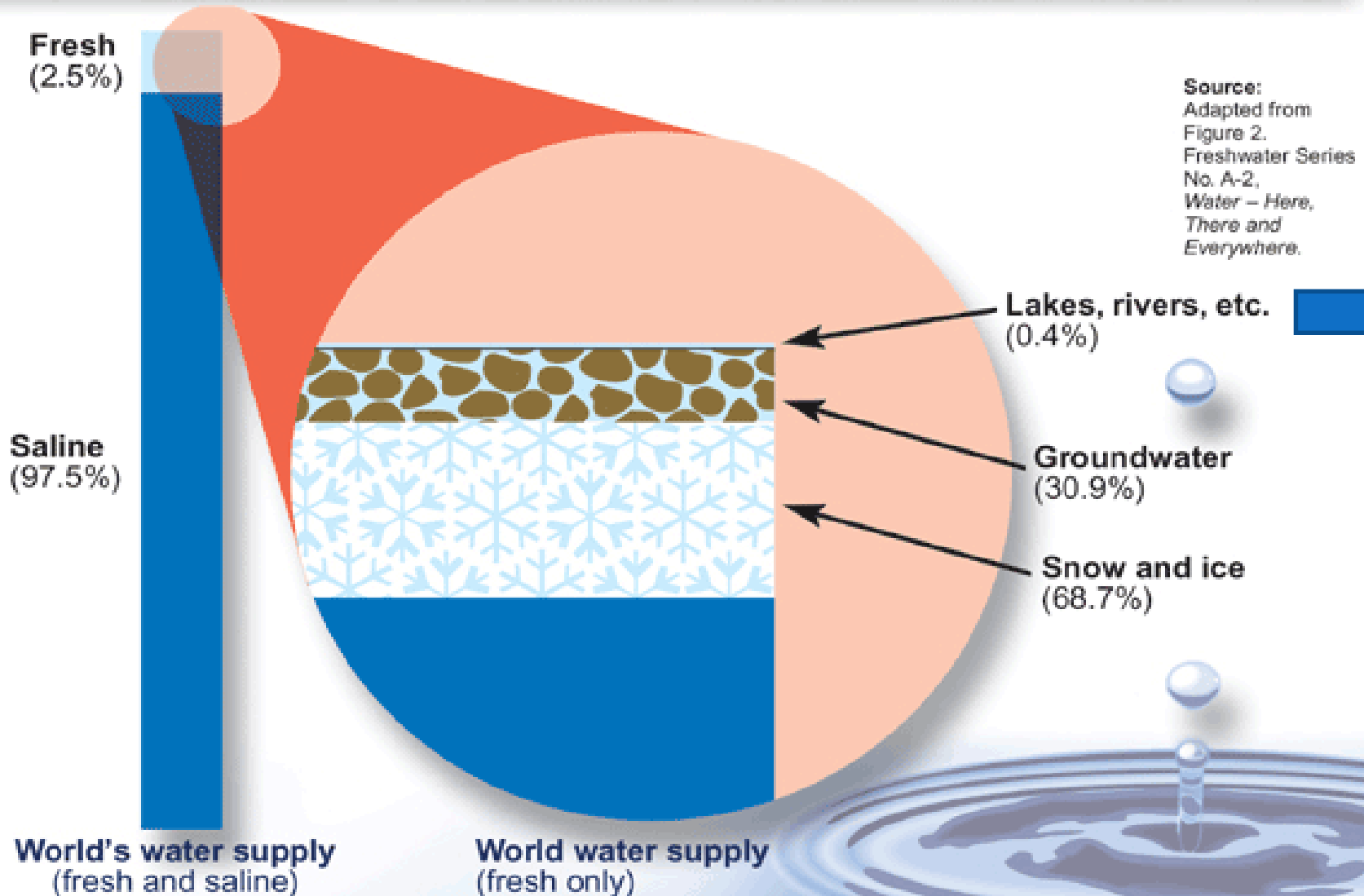
HYDROLOGY V/S OTHER SCIENCES

- **Hydrology is an inter disciplinary science which requires the knowledge of:**
 - **Climatology and meteorology (precipitation)**
 - **Soil science (infiltration)**
 - **Geology (groundwater)**
 - **Geomorphology (surface runoff)**
 - **Fluid mechanics (stream / groundwater flow)**
 - **Chemistry & physics (Physico-chemical properties)**
 - **Biology (health hazards), etc....**

IMPORTANCE OF HYDROLOGICAL STUDIES

- The study of processes such as precipitation, evapotranspiration, runoff and their interaction
- Estimation of water resources
- Management and conservation of water on global and regional scales
- The study of problems such as floods and droughts and strategies to combat them

Groundwater and the world's freshwater supply



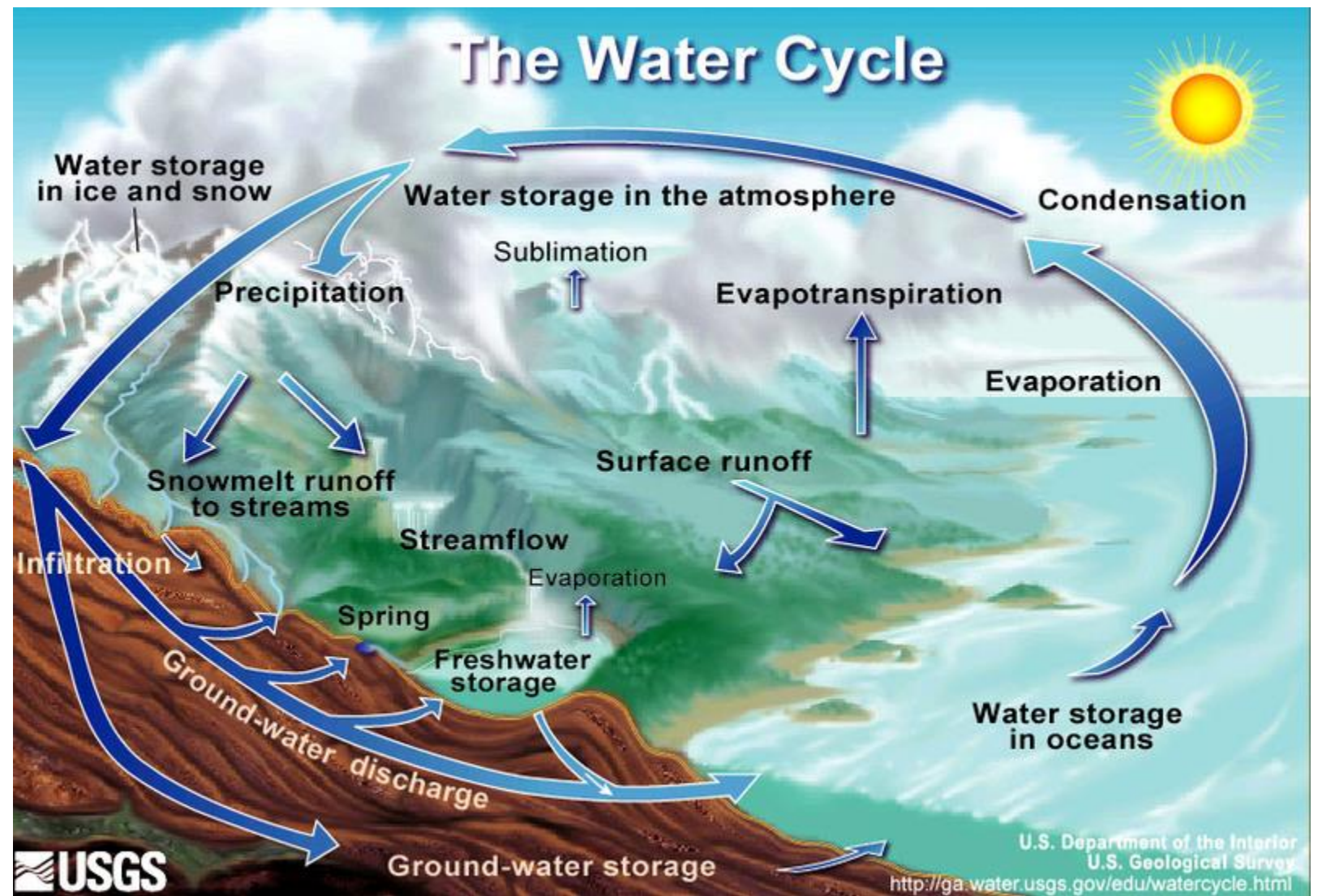
Storage in the atmosphere = 0.001%

Water vapor, Clouds,
condensed water vapor

HYDROLOGIC CYCLE...

- Water, irrespective of different states, involves dynamic aspect in nature.
- The dynamic nature of water, the existence of water in various state with different hydrological process result in a very important natural phenomenon called **Hydrologic cycle.**

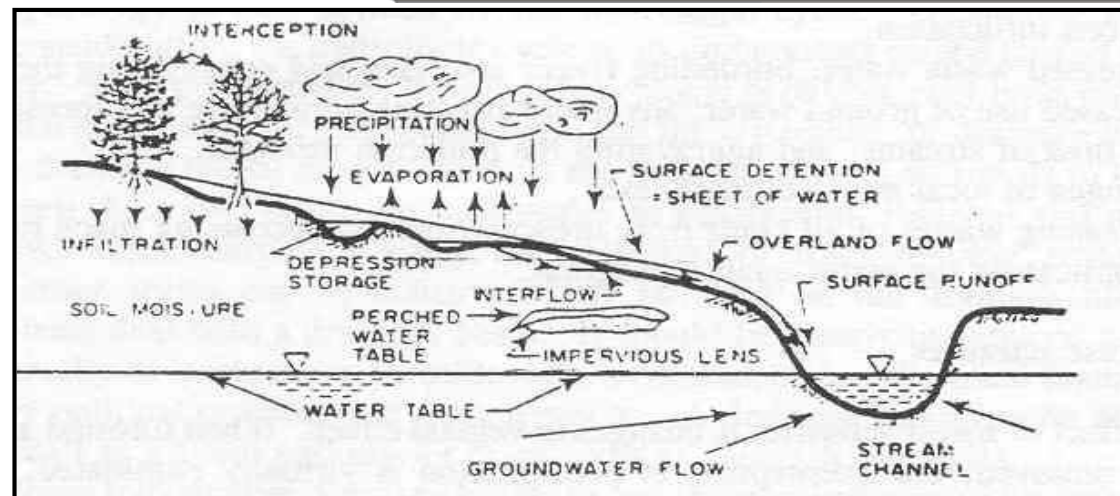
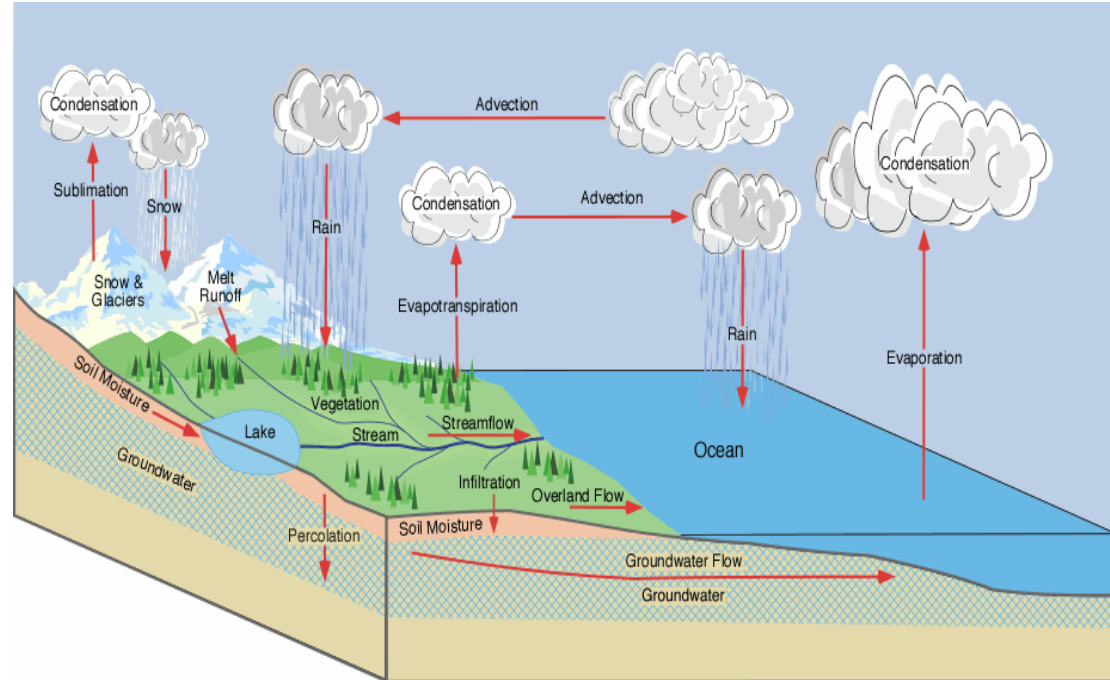
The water cycle (known scientifically as the **hydrologic cycle**) refers to the continuous exchange of water within the hydrosphere, between the atmosphere, soil water, surface water, groundwater, and plants. Water can change states among liquid, vapor, and ice at various places in the water cycle.



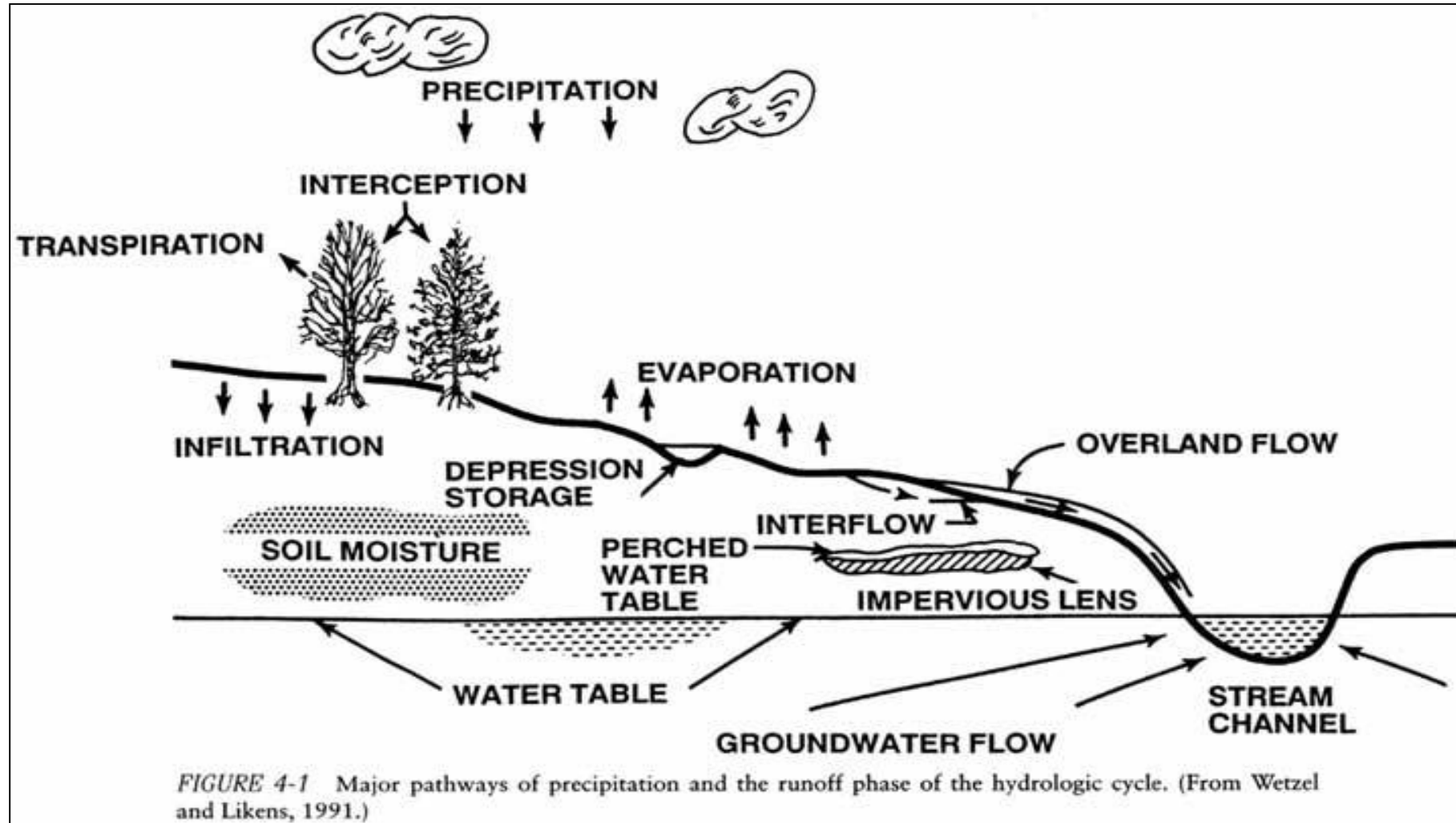
Evaporation of water from water bodies, such as oceans and lakes, formation and movement of clouds, rain and snowfall, stream flow and ground water movement are some examples of the dynamic aspects of water.

PROCESSES IN TERRESTRIAL HYDROLOGICAL CYCLE

- **Precipitation**
- **Condensation**
- **Evaporation**
- **Sublimation**
- **Interception**
- **Evapotranspiration**
- **Snowmelt**
- **Runoff (Stream flow)**
- **Surface storage**
- **Infiltration**
- **Deep Percolation**
- **Subsurface Flow**
- **Advection**
- **Dispersion**



A Simple overview of the hydrologic cycle



The Water Cycle (in detail)

- The volume (M) of water at the surface of the Earth is enormous: $1.37 \cdot 10^9 \text{ km}^3$! (total reservoir) – The Oceans cover 71% of the Earth's surface (29% for the continent masses above sea level)

Reservoir	Volume (km^3)	% Total
Biosphere	$0.6 \cdot 10^3$	0.00004
Rivers	$1.7 \cdot 10^3$	0.0001
Atmosphere	$13 \cdot 10^3$	0.001
Lakes	$125 \cdot 10^3$	0.01
Groundwater	$9500 \cdot 10^3$	0.68
Glacial and other land ice (?)	$29000 \cdot 10^3$	2.05
Oceanic water and sea ice	$1,370,000 \cdot 10^3$	97.25
Total	$1,408,640 \cdot 10^3$	100

Adapted from Berner & Berner (*The Global Water Cycle*; Prentice Hall, 1987)



WATER RESOURCES OF INDIA

Annual Water Availability in India

Total precipitation	4000 BCM
Annual water availability (after accounting for losses in the form of evaporation etc.)	1869 BCM
Utilizable water (the available water cannot be fully utilized due to topographical and hydrological constraints and the need for allowing certain amount of water to flow in the river for maintaining the river regime.)	1123 BCM
- Surface water - Ground water	690 BCM 433 BCM

India's Land Resources

Geographical Area	329 mha
Non-cultivated Area Barren/Waste Land	7% 23%
Cultivated Area (CA)	47%
Irrigated Area (produces 55%)	37% of CA
Rainfed Area (produces 45%)	63% of CA

India – A Comparison with World

India's Land Resources	2% of the World
India's Freshwater Resource	4% of the World
India's Cattle Population	10-15% of World

India's Water Demands for Various Uses

Use	Year 2010		Year 2025		Year 2050	
	Water Demand (BCM)	% of total demand	Projected Demand (BCM)	% of total demand	Projected Demand (BCM)	% of total demand
Irrigation	557	78%	611	72%	807	69%
Domestic	43	6%	62	7%	111	9%
Industries	37	5%	67	8%	81	7%
Environment	5	1%	10	1%	20	2%
Others	68	10%	93	12%	161	13%
Total	710	100%	843	100%	1180	100%

**Per Capita
Availability <
1700 Cubic
Meter per year**

**• Water
Stressed**

**Per Capita
Availability <
1000 Cubic
Meter per year**

**• Water
Scarce**

Per Capita Water Availability in India With Time

Year	Population (in millions)	Per Capita annual water availability (in cubic meter)
1951	361	5177
2001	1027	1820
2025 (projected)	1394	1341
2050 (projected)	1640	1140

Key Water Resources Issues in India

Spatial & temporal variability of rainfall

Low Per capita water availability

Flood & drought management

Conjunctive management of surface & groundwater

Rainwater harvesting and watershed management

Inter-basin water transfer

Hydrological information system

Key Water Resources Issues in India..

Water quality conservation & environment restoration

Recycle and reuse of water

Urban drainage and flood management

Vulnerability and adaptation to climate change

Water rights, ownership, and pricing

Desalination and water management in coastal areas

Development of DSS for optimum use at basin scale

Capacity building, people's participation & mass awareness

What is Groundwater?

- **Groundwater** is water below the water table where all of the pore spaces are filled with water.
- The area above the water table where the pore spaces are only partially filled with water is called the **Unsaturated Zone**.

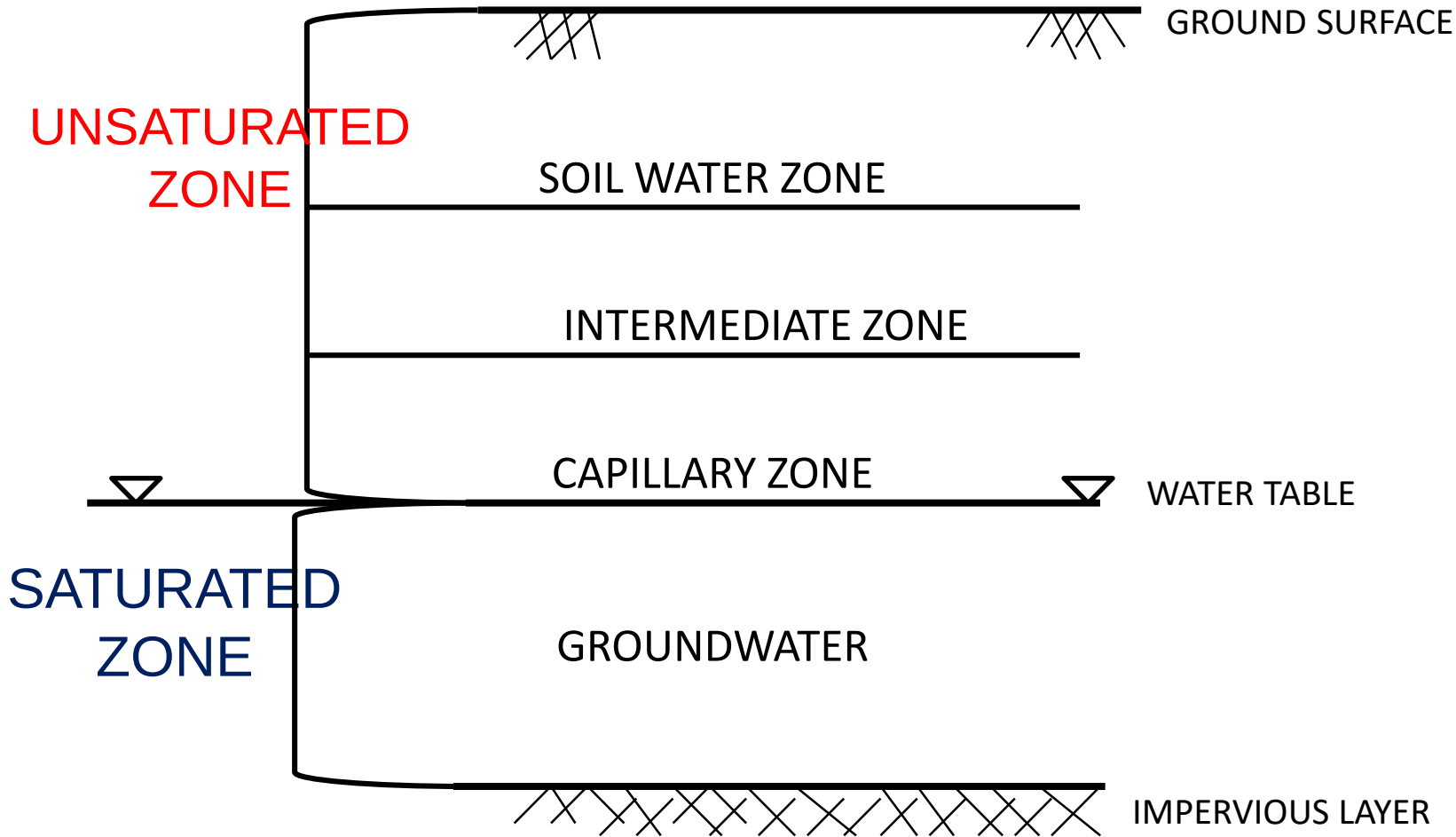
Subsurface Water
(Water beneath
Earth surface)

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graph TD; A["Subsurface Water  
(Water beneath  
Earth surface)"] --> B["Saturated"]; A --> C["Unsaturated"];
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Saturated

Unsaturated

Vertical Distribution of Groundwater

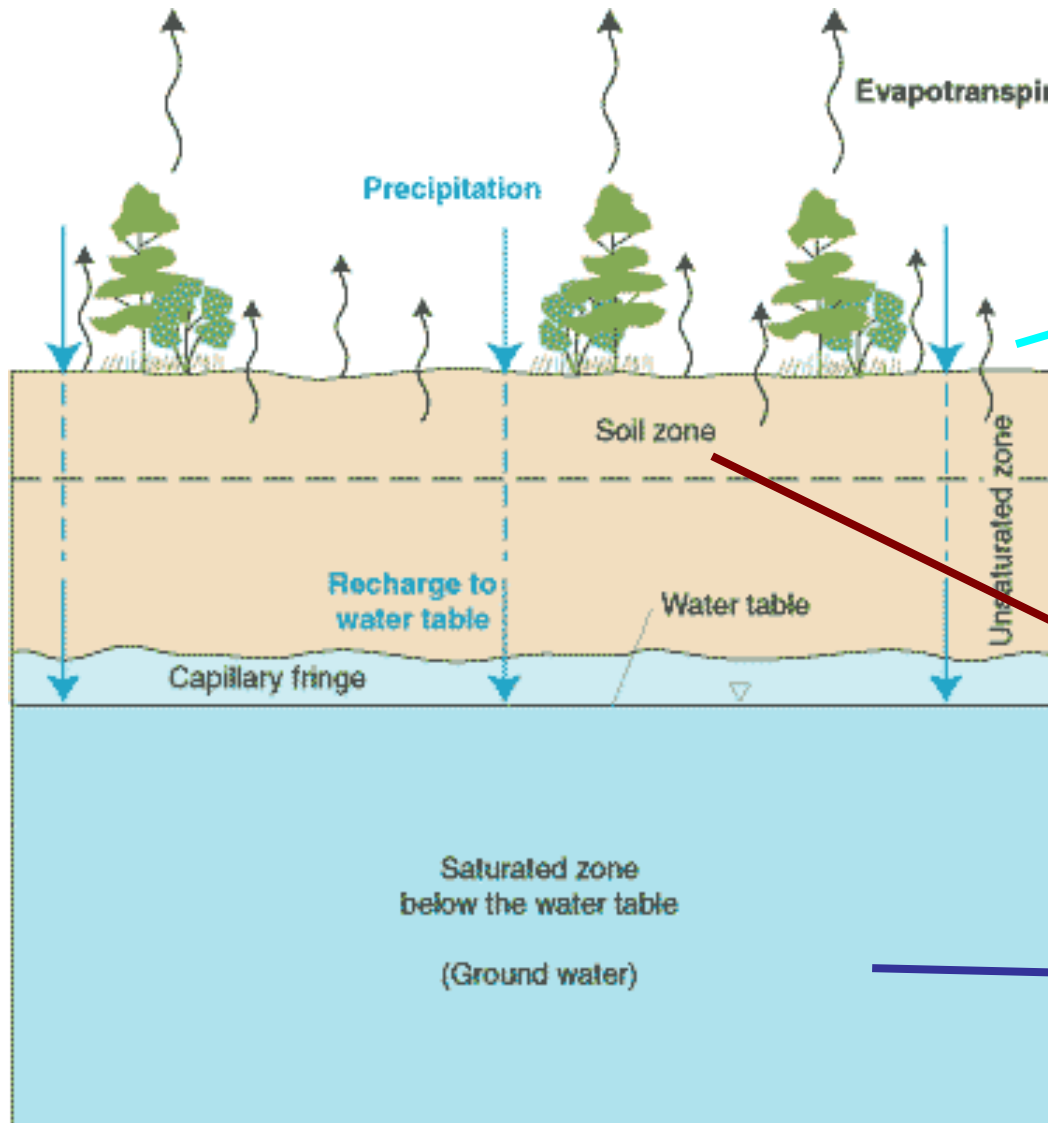


The soil zone extends from the land surface to maximum depth of a meter or two and is the zone that supports Plant growth. It is crisscrossed by living roots, by voids left by decayed roots and animal/insects burrows.

The soil zone is underlain by intermediate zone, which differ in thickness from place to place

The lowest part of the unsaturated zone is occupied by the *capillary fringe*, the subzone between the unsaturated and saturated zones. The capillary fringe results from the attraction between water and rocks. As a result of this attraction, water clings as a film on the surface of rock particles and rises in small-diameter pores against the pull of gravity. Water in the capillary fringe and in the overlying part of the unsaturated zone is under a negative hydraulic pressure—that is, it is under a pressure less than the atmospheric (barometric) pressure. The water table is the level in the saturated zone at which the hydraulic pressure is equal to atmospheric pressure and is represented by the water level in unused wells. Below the water table, the hydraulic pressure increases with increasing depth.

Description of unsaturated and saturated Zone



Soil
Moisture



Ground water